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The influence of massive early merger events on the chemo-dynamics of galaxy discs

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ABSTRACT
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Key words:
Galaxy: structure — Galaxy: evolution — Galaxy: kinematics and dynamics — galaxies: formation — Galaxy: disk — methods: numerical

1 INTRODUCTION

2 METHODS

3 RESULTS

3.1 some different elements [X/Fe] vs. [Fe/H]

do different elements give us insight into different formation epochs of the MW? detailed galactic archeology is left for future work.

3.2 relative contribution of different enrichment channels in L* galaxies: SNII vs. AGB vs. SNIa

4 CONCLUSION

Our results are summarized as follows:

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APPENDIX A: ROTATION VELOCITY OF STARS

APPENDIX B: OXYGEN ABUNDANCE DISTRIBUTION

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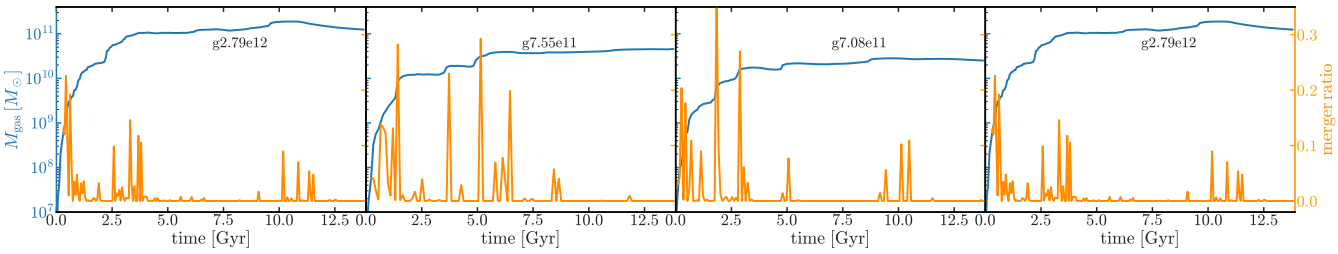

Figure 1. Gas mass (blue line, left axis) and merger ratio (orange line, right axis) as a function of time. We identify several gas rich mergers that contribute around 10% of gas to the main galaxy at various times of 2-4 Gyr and a late time merger starting around 9.5 Gyr that contributes 10% of gas and continues for several pericenter passages and goes on for ~ 2 Gyr.

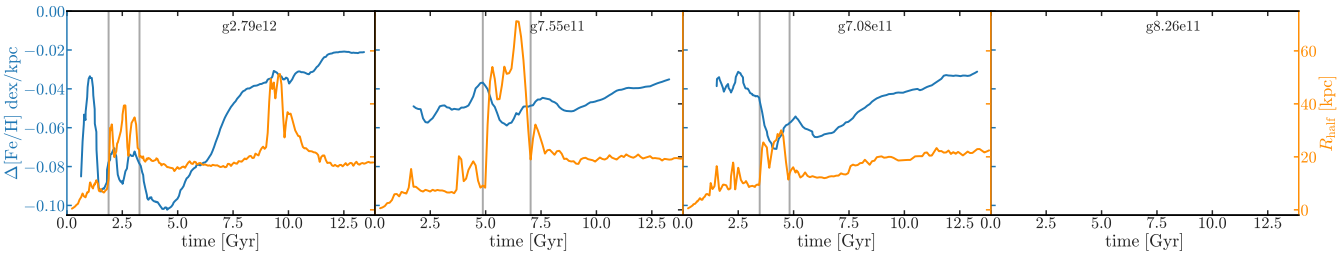

Figure 2. Half mass radius of the cold gas (orange line, right axis) and the metallicity gradient (blue line, left axis) as a function of time. The sharp increases in the cold gas size around the times of gas rich mergers coincide with a sudden steepening of the metallicity gradient of ~ 0.02 dex/kpc. The addition of relatively unenriched cold gas by the merging satellite leads to a sudden increase of the star-forming cold gas mass in the outskirts of the galaxies while the central parts are unaffected. This steepens the gradient.

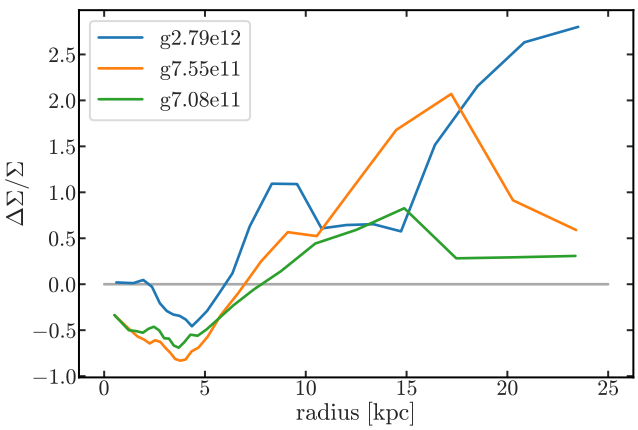

Figure 3. Relative change in cold gas surface density profile before and after the merger events. Outside of a radius of $\sim 5-10$ kpc the merging satellites contribute significant amounts of cold gas up to 200%.

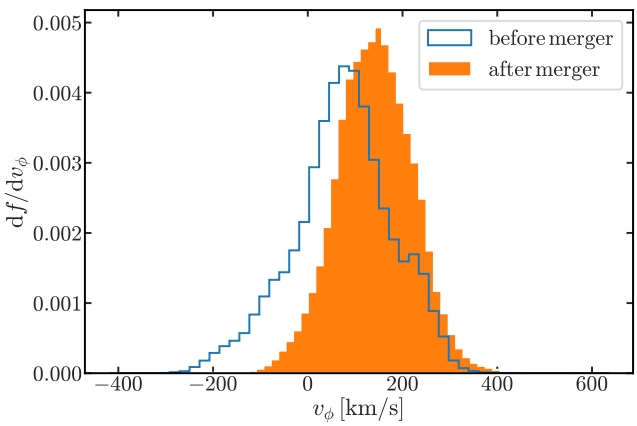

Figure 4. Histogram of the cold gas velocity in the disk plane, v_ϕ, before and after the merger. The peak of the v_ϕ distribution shifts from 73 km/s before the merger to 140 km/s after the merger, almost doubling the rotation speed of the cold gas disk.

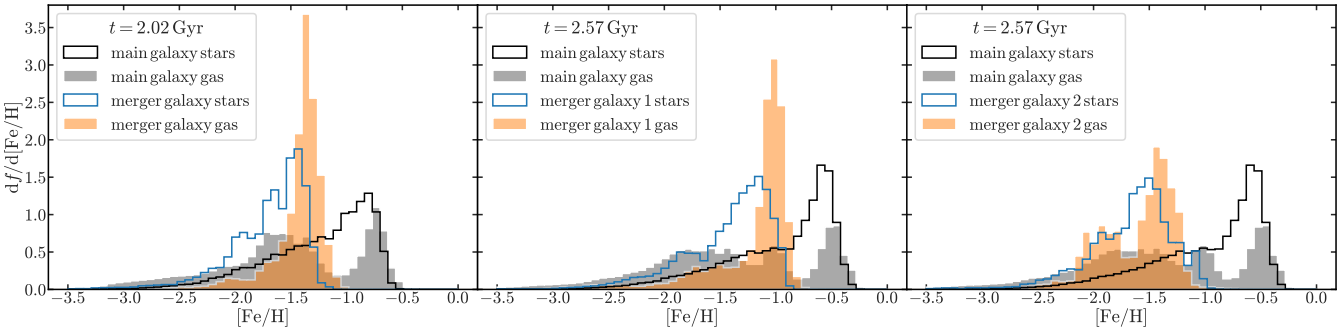

Figure 5. Metallicity distribution function of the main galaxy and the merging satellite shortly before coalescence for all three merger events. Filled histograms show the gaseous MDF while steps show the MDF for stars. In any case, the gas metallicity of the satellite is $\sim 0.5 - 0.75$ dex lower that the main galaxy's gas metallicity.

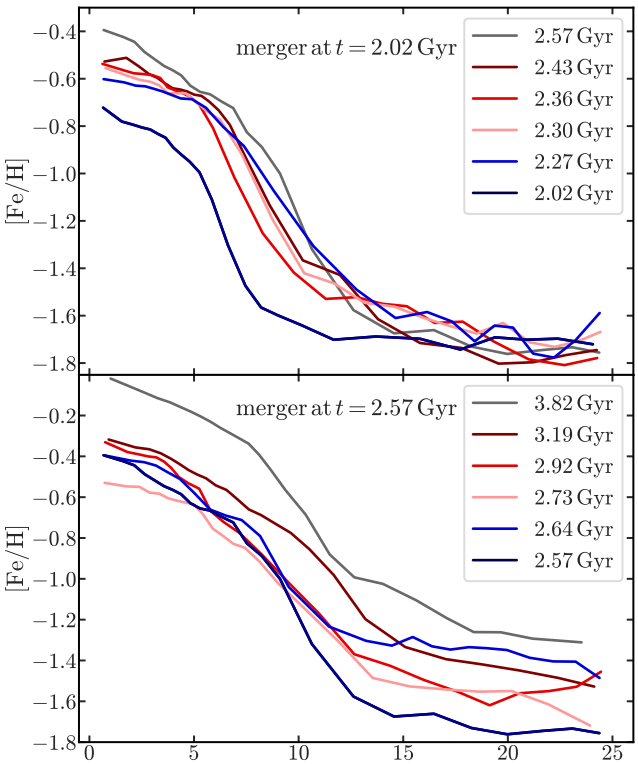

Figure 6. Metallicity gradient of the cold gas disk during the merger events. We find that the contribution of additional metal poor gas lowers the metallicity in the outskirts of the disk effectively steepening the metallicity gradient by ~ 0.1 dex.

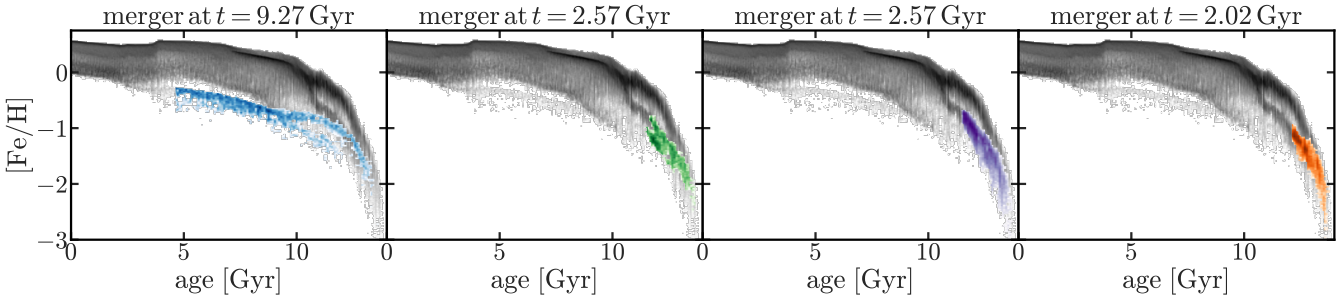

Figure 7. Age-metallicity relation.

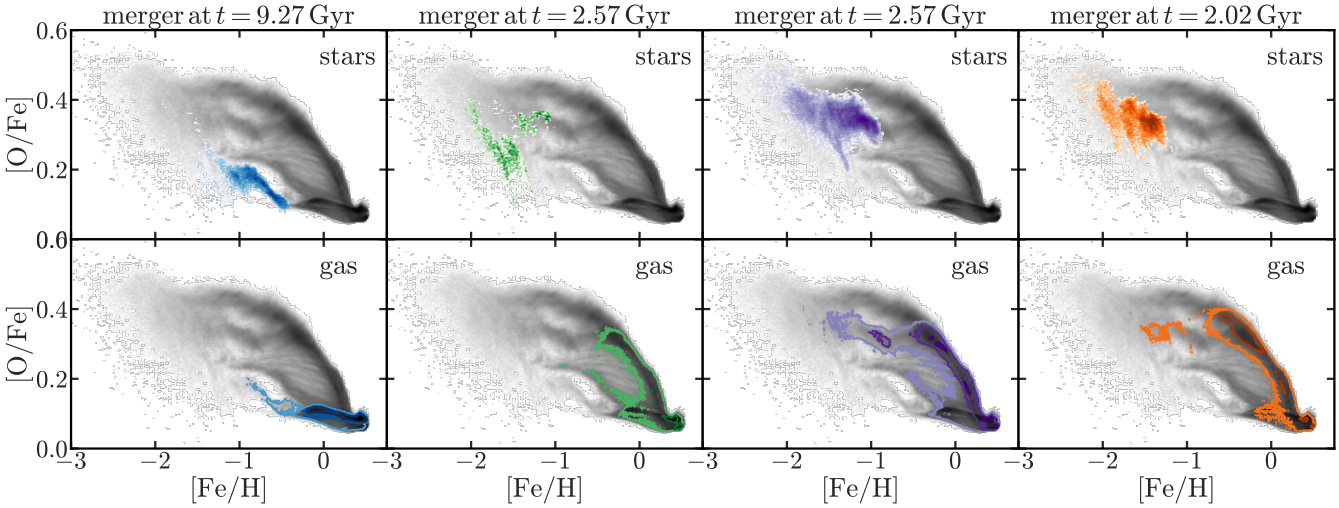

Figure 8. Oxygen vs. metallicity.

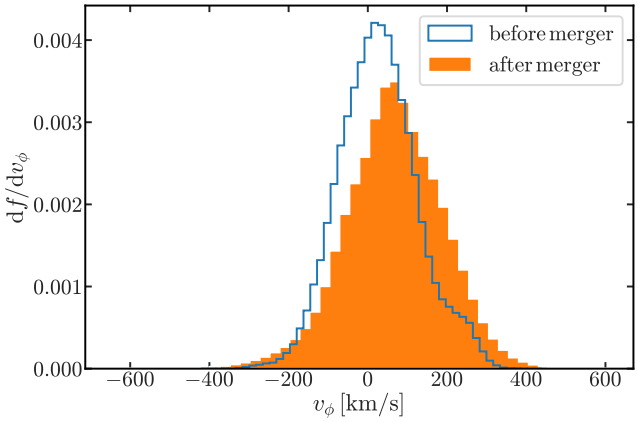

Figure A1. Same as Fig. 4 but for the stellar rotation velocity in the disk plane, v_ϕ, before and after the merger. The peak of the v_ϕ distribution shifts from 27 km/s before the merger to 66 km/s after the merger, more than doubling the rotation speed of the stellar disk.

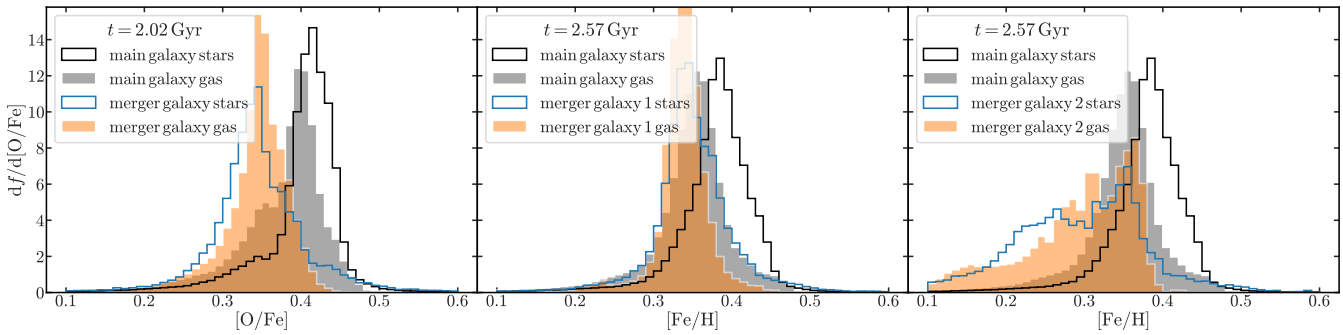

Figure B1. Same as Fig. 5 but for the oxygen abundance. Filled histograms show the gaseous oxygen abundance distribution while steps show the one for stars. The gas oxygen abundance of the satellite is ~ 0.1 dex lower that the main galaxy's gas oxygen abundance.